

A DATA SCIENCE LAB REPORT ON WEB SCRAPING AND DATASET PREPARATION

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1. Introduction

Web scraping is an essential data-gathering technique used in data science and software engineering to extract structured information from unstructured web pages. This lab project focuses on scraping e-commerce book data from a practice website, "Books to Scrape" (books.toscrape.com). The primary purpose of this lab is to understand HTML structure parsing, handle pagination, clean raw web data, and store it into a structured format (e.g., CSV/JSON) for further analytical use.

2. Domain Description

The target domain is an online book store interface that mimics a real-world e-commerce platform. It contains multiple book categories, individual book listings, pricing details, stock availability, star ratings, and detailed descriptions. Since this website is designed specifically for web scraping practice, it provides a stable environment with standard HTML elements without encountering aggressive anti-scraping measures (such as IP blocking or Captchas).

3. Objectives

To automatically fetch web content using HTTP requests.

To parse HTML DOM elements using appropriate scraping libraries.

To extract key data attributes: Book Title, Price, Rating, Availability Status, and Product URL.

To handle pagination across multiple pages to scrape a complete dataset.

To perform basic data cleaning and store the dataset in a structured CSV file.

4. Data Source

Source Name: Books to Scrape

URL: [\[http://books.toscrape.com/\]\(http://books.toscrape.com/\)](http://books.toscrape.com/)

Data Fields Extracted:

Title: Full name of the book.

Price: Cost of the book in British Pounds (£).

Rating: Star rating (One to Five).

Availability: In-stock status.

Product Link: Direct URL to the book's detail page.

5. Methodology

The data extraction pipeline followed a structured programmatic approach:

1 HTTP Request Execution: Sent HTTP GET requests to the target pages using Python's requests module (or equivalent library).

2 HTML Parsing: Processed the raw HTML document tree using BeautifulSoup (or Scrapy / Selenium) to locate specific tags (e.g., `<article class="product_pod">`).

3 Attribute Extraction: Loop through HTML cards to extract target data:

Extracted text from `<h3>` tags for Titles.

Extracted CSS class names to determine Star Ratings.

Captured text nodes for Prices and Availability.

4 Pagination Handling: Programmatically identified the "Next" button link (`<li class="next">`) and recursively scraped consecutive pages until all pages were processed.

5 Data Cleaning & Storage: Stripped unnecessary characters (e.g., currency symbols, whitespace) and formatted the structured output into a standard CSV format using Pandas.

6. Conclusion

This lab successfully demonstrated the automated collection of structured web data from an online book directory. The implemented web scraper efficiently traversed multiple pages, parsed relevant HTML components, and generated a clean dataset ready for data analysis or machine learning tasks. This exercise highlights the importance of understanding DOM hierarchy and request management in web data extraction projects.